VISVESVARAYA TECHNOLOGICAL UNIVERSITY

“Jnana Sangama”, Belagavi– 590 018

A Mini Project Report on

“Gas Leak Detection Using NODEMCU”

Submitted in the partial fulfilment of the requirements for the award of degree of

BACHELOR DEGREE

IN

ELECTRONICS & COMMUICATION ENGINEERING

Submitted by

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CERTIFICATE

This is to Certify that **Mr. SHASHANK BEVANUR, Mr. SHIVARAJ P BIRADAR, Mr. SHIVARAJ U ANANTAPUR & Mr. VINAY P BONGALE** Bonafide students of **V. P. Dr. P. G. Halakatti College of Engineering & Technology**, has completed the Mini project work entitled **"Gas Leak Detection Using NodeMCU"** in partial fulfilment for the award of **Bachelor Degree in Electronics & Communication Engineering** of the **Visvesvaraya Technological University, Belagavi** during the year **2025-2026**. It is certified that all corrections/suggestions indicated for Internal Assessment been incorporated in the report deposited in the departmental library. The Mini project report has been approved as it satisfies the academic requirements in respect of project work prescribed for the said degree.

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DECLARATION

We, **Mr. SHASHANK BEVANUR , Mr. SHIVARAJ P BIRADAR, Mr. SHIVARAJ U ANANTAPUR and Mr. VINAY P BONGALE,** a students of Third year B.E., Electronics and Communication Engineering, BLDEA's V.P. Dr. P.G. Halakatti College of Engineering and Technology, here by declare that the Mini Project work titled "Gas Leak Detection Using NodeMCU" has been carried out by us and submitted in partial fulfillment of the course requirement for the award of degree in Bachelor of Engineering in Electronics and Communication of Visvesvaraya Technological University, Belagavi during the academic year 2025-2026. We also declare that we have not submitted this Mini project report to any other university for the award of the degree.

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ABSTRACT

Gas leakages pose significant threats to life, property, and the environment, especially in residential, industrial, and commercial areas. Early detection of such leaks is critical for preventing accidents and ensuring safety. This project proposes a cost-effective and efficient gas leak detection system using the NodeMCU ESP8266 microcontroller integrated with an MQ-series gas sensor (e.g., MQ-2 or MQ-135). The system continuously monitors the concentration of combustible gases such as LPG, methane, and propane. When the gas concentration exceeds a predefined threshold, the system triggers an alert by activating a buzzer and sending a real-time notification to the user via Wi-Fi using platforms like Blynk or IFTTT. The NodeMCU's built-in Wi-Fi capability enables remote monitoring and data logging, enhancing the system's usability and scalability. This solution offers a smart, low-cost, and IoT-enabled approach to enhance safety in gas-prone environments.

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CHAPTER 1

INTRODUCTION

Introduction:

Gas leak detection is crucial for safety in various environments, such as homes, industrial settings, and laboratories. Using an NodeMCU for this purpose can be both an educational and practical approach. Here’s a brief introduction to how you can set up a gas leak detection system with NodeMCU. In this project, we will build Gas Leak Detection & Alarm System using MQ2 Gas Sensor and NodeMCU (ESP8266) Board. The MQ2 sensor is a versatile gas sensor capable of detecting a wide range of gases as well as smoke. This initiative is centered on creating a mock-up of a gas leak detection system utilizing an NodeMCU as the core controller. It employs an MQ2 gas sensor alongside an RGB LED to monitor gas levels persistently. When the detected gas levels exceed a certain preset limit, an alert is triggered via a buzzer, and the RGB LED glows red to indicate danger. If the gas levels are within safe limits, below the set threshold, the system keeps the alarm silent and the LED displays a green light, signaling safety. **[3]**

CHAPTER 2

METHODOLOGY

NodeMCU is an open-source development board that uses Lua programming and is mainly used for IoT applications. It runs on the ESP8266 Wi-Fi chip made by Espressif Systems, and its hardware is based on the ESP-12 module. The project methodology is shown in Figure 2.1 below.**[1]**

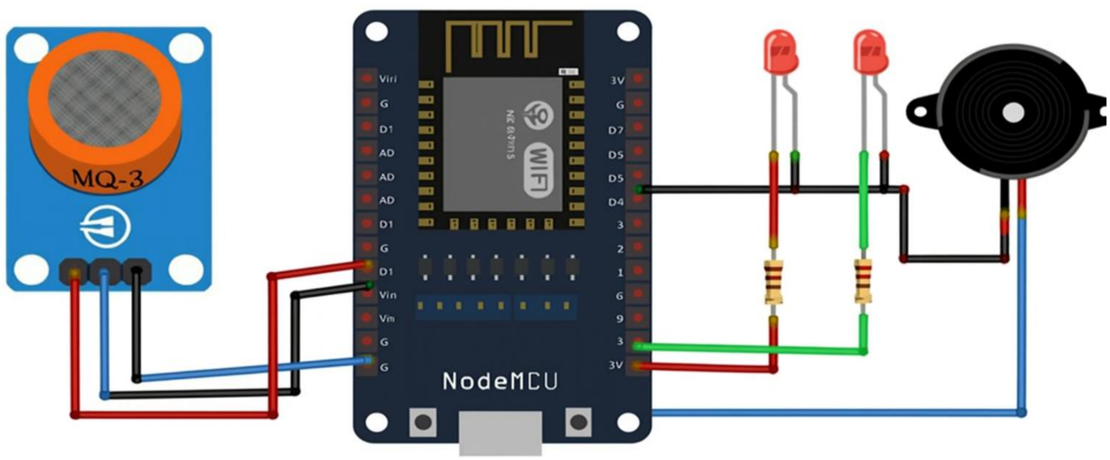

Fig 2.1: Block diagram for Gas leak detection using NodeMCU

1. **Objectives of the Experiment**:

   Identify Target Gases: Determine which gases need to be detected (e.g., methane, propane, carbon monoxide).

   Set Safety Thresholds: Define the concentration levels at which alerts should be triggered.

   Determine Output Mechanisms: Decide how you want to alert users (e.g., buzzer, LED, display, SMS).

2. **Components**:

   Gas Sensor:

   Select the Right Sensor: Choose a gas sensor that is appropriate for the gases you need to detect. For example, MQ-2 for methane and propane, MQ-7 for carbon monoxide.

   Understand Sensor Specifications: Read the datasheet to understand the sensor's operating voltage, output type (analog or digital), and calibration requirements.

NodeMCU Board:

Select an NodeMCU Model: Use a board with sufficient analog/digital pins (e.g., ESP8266 or ESP32).

Output Devices:

Buzzer and LED: For audible and visual alerts.

Exhaust fan: For exhausting air

3. Component Connections:

Gas Sensor:

VCC to NodeMCU 5V (or 3.3V if required). GND

to NodeMCU GND.

Analog Output (AO) to an analog input pin (e.g., A0) for analog sensors.

Digital Output (DO) to a digital pin (optional, for sensors with a built-in comparator).

Buzzer:

Connect one wire to a digital pin (e.g., D2) and the other to GND. Exhaust

Fan: connect one wire to fan and another to relay

LED:

Connect the anode (long leg) to a digital pin (e.g., D3) through a resistor (220Ω to 1kΩ), and the cathode (short leg) to GND.

CHAPTER 3

HARDWARE REQUIRMENTS

1. **MQ-2 Gas sensor**

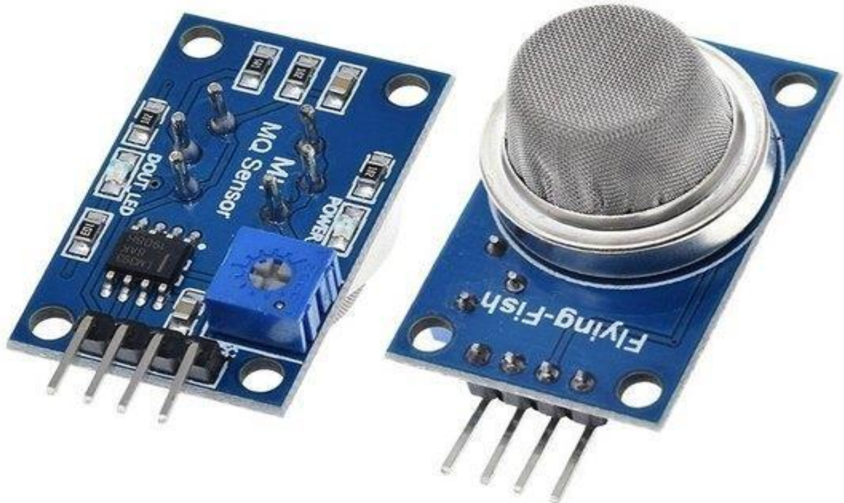

Fig.3.1 MQ-2 Gas sensor

The MQ-2 gas sensor is a versatile and cost-effective solution for detecting a variety of gases. The MQ-2 sensor stands out as a multifunctional gas detector that can identify various gases such as alcohol, carbon monoxide, hydrogen, isobutene, liquefied petroleum gas, methane, propane, and smoke. Its affordability and user-friendly attributes make it a favorite choice for beginners in the field.[4]

Design and Functionality

The MQ-2 sensor consists of a small tin dioxide (SnO_2) layer on an aluminum oxide tube. The SnO_2 is a semiconductor material that has lower conductivity in clean air. When the target gas is present in the environment, the conductivity of the SnO_2 increases as the gas reacts with the oxygen on the surface of the sensor. This change in conductivity is used to measure the gas concentration. The sensor has a built-in potentiometer to adjust the sensitivity of the sensor to the presence of different gases. By adjusting the load resistance of the sensor, you can change the type of gas to which it is most sensitive.

2. NodeMCU (ESP8266)

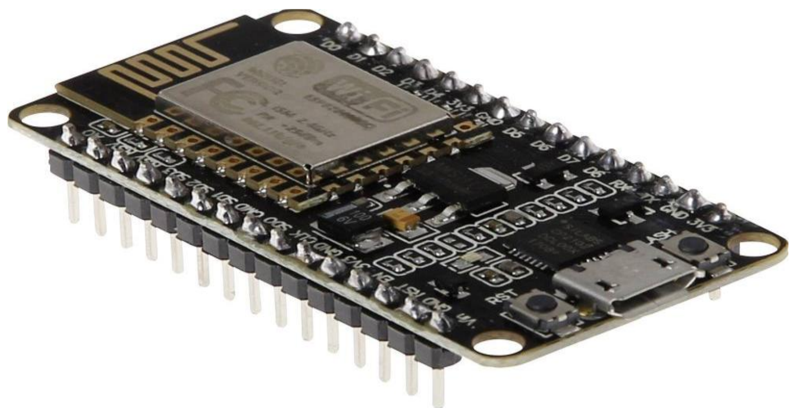

Fig 3.2 NodeMCU (ESP8266)

NodeMCU (ESP8266) is a microcontroller-based development board designed for building IoT (Internet of Things) applications **[1]**. It is built around the **ESP8266 Wi-Fi module**, which provides built-in wireless networking capabilities, making it ideal for projects that require internet connectivity. Like Arduino, NodeMCU includes both **hardware** (the development board) and **software** (the ESP8266/Arduino IDE environment) used to write, upload, and run code.

The board features a powerful microcontroller that can run user-written programs to control sensors, actuators, and various electronic components. NodeMCU provides multiple **digital I/O pins**, an **analog input pin**, and **power pins**, allowing easy interfacing with external modules such as gas sensors, buzzers, LEDs, and relays. It supports programming through the **Arduino IDE, Lua scripts, or PlatformIO**, giving developers flexibility in how they write their code.

NodeMCU is powered through a **micro-USB connection**, which is also used to upload programs. Because the board includes built-in Wi-Fi, it can connect to cloud platforms, send sensor data online, or enable remote control of devices—all without external networking hardware. Both the hardware design and much of the supporting software are open-source, allowing customization and community-driven improvements.

Interfacing an **MQ-2 gas sensor with a NodeMCU** is simple, involving basic wiring and programming to read gas levels, trigger alarms, or send data wirelessly to an IoT platform.

3. Buzzer

Fig 3.3 Buzzer

A buzzer is an electronic component designed to produce sound, often used in various devices to provide audible alerts or notifications. It operates by converting electrical energy into acoustic signals, utilizing either electromechanical mechanisms or piezoelectric effects. Electromechanical buzzers use a vibrating diaphragm or reed activated by an electromagnetic coil to generate sound, while piezoelectric buzzers rely on the piezoelectric effect, where a material vibrates when an electrical voltage is applied, creating sound waves. Buzzers come in two main types: active and passive. Active buzzers contain an internal oscillator and emit a fixed tone when powered, making them straightforward to use. In contrast, passive buzzers require an external AC signal to produce sound and offer greater flexibility in tone generation. Commonly used in alarm systems, electronic devices, automotive applications, and industrial equipment, buzzers are valued for their ability to provide clear, attention-grabbing notifications. Their versatility, coupled with low power consumption and ease of integration with various systems, makes them an essential component in many electronic projects.

4. 1-channel relay

Fig 3.4 1-channel relay

A single-channel relay is an electronic component used to control high-power or high-voltage devices through a low-power control signal. Essentially, it acts as an electrically operated switch that can turn a circuit on or off based on an input signal. The relay consists of a coil, which, when energized by a control signal, creates a magnetic field that moves a switch mechanism to either open or close a circuit. This allows a low-current control signal, such as that from a microcontroller or sensor, to manage larger loads like motors, lamps, or appliances safely and efficiently. Single-channel relays are widely utilized in automation systems, home appliances, and various electronic projects where isolation between control and power circuits is essential. Their primary advantage lies in their ability to handle higher current and voltage levels than the control signal alone could manage, making them crucial for interfacing low-power electronics with high-power applications.

5. DC Fan

Fig 3.5 DC Fan

A DC fan is an electrical device that uses direct current (DC) to generate airflow, typically for cooling or ventilation purposes. Unlike AC fans, which operate on alternating current (AC), DC fans are powered by a DC voltage source, such as batteries or DC power adapters. These fans consist of a motor that drives blades to move air, creating a cooling effect or improving ventilation in various environments. DC fans are valued for their efficiency, as they generally consume less power compared to their AC counterparts and can be controlled more precisely with electronic circuits. They are commonly used in computer cooling systems, automotive applications, and small household appliances. Additionally, DC fans offer advantages such as quieter operation and the ability to be easily regulated via pulse-width modulation (PWM) for variable speed control. Their compact size and energy efficiency make them suitable for a wide range of applications, from consumer electronics to industrial machinery.

6. Servo motor

Fig 3.6 Servo Motor

A servo motor is a specialized type of motor designed for precise control of angular position, velocity, and acceleration. Unlike standard motors that run continuously at a set speed, a servo motor is equipped with a feedback mechanism, typically an encoder or potentiometer, that allows for exact positioning and fine adjustments. It operates based on a control signal that dictates its position, and it adjusts its output accordingly to maintain the desired angle. This makes servo motors ideal for applications requiring high accuracy and repeatability, such as in robotics, automated manufacturing systems, and precision instruments. They are commonly used in steering mechanisms, camera focus systems, and various automated control systems. The ability to control the motor's position with high precision and stability, combined with its relatively compact size, makes the servo motor a versatile component in both hobbyist projects and advanced industrial applications.

CHAPTER 4

4.1 RESULT AND DISCUSSION

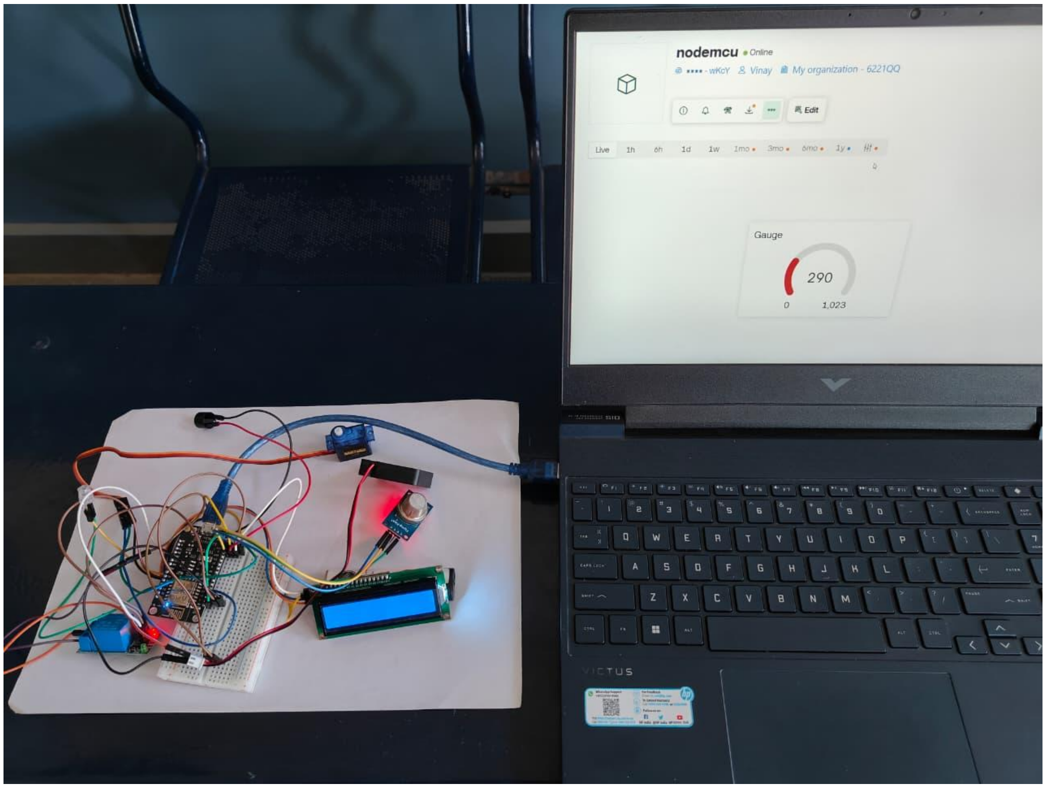

Fig 4.1 Model

The circuit is ready for testing once the code uploading is done. When the gas level is lower, the green LED turns ON and no alarm sound is produced. This means, if the value is below 400, the buzzer is silenced using no Tone(), and the LED turns green. This means if the value exceeds 400, the buzzer is activated using tone(), and the RGB LED turns red. If the concentration drops below the threshold, the alarm is deactivated and the LED switches to green, indicating a safe environment. It's crucial to note that this demo is purely illustrative and shouldn't replace real gas leak detection systems.

You can make a better version of this project which are as follows:

Gas Leakage Detection with SMS Alert

Gas Leakage Detector with Email Alert Notification

CHAPTER 5

ADVANTAGES AND DISADVANTAGES

5.1 Advantages

1. Affordability

Cost-Effective Components: Both the MQ-2 gas sensor and NodeMCU board are relatively inexpensive, making it a budget-friendly solution for gas detection and alarm systems. This cost-effectiveness allows for wider adoption in both personal and professional settings.

2. Real-Time Monitoring

Immediate Detection: The system continuously monitors gas concentrations and provides real-time data. This allows for prompt detection of gas leaks, enabling quick responses to potential hazards and minimizing the risk of accidents.

3. Versatility

Broad Gas Detection: The MQ-2 sensor is capable of detecting various gases, including methane, propane, and carbon monoxide. This versatility makes it suitable for a range of applications, from household safety to industrial environments.

4. Clear Alert Mechanism

Visual and Audible Alarms: The integration of a buzzer and LED provides both visual and audible alerts when gas concentrations exceed predefined thresholds. This ensures that users are immediately informed of potential leaks, enhancing overall safety.

5. Ease of Implementation

User-Friendly Development: The NodeMCU platform is known for its simplicity and ease of use. With straightforward programming and a supportive community, implementing and customizing the gas detection system is relatively easy, even for those with limited experience in electronics.

6. Wi-Fi Connectivity

Remote Monitoring Capability: The built-in Wi-Fi module of the NodeMCU allows the system to send alerts, upload data to cloud platforms, and enable remote supervision through smartphones or computers. This enhances convenience and expands the system's functionality beyond local detection. **[2]**

5.2 Disadvantages

1. Sensitivity and Calibration Issues

    Calibration Required: The MQ-2 sensor requires periodic calibration to maintain accuracy. Variations in environmental conditions, such as humidity and temperature, can affect its sensitivity and performance, necessitating regular adjustments.

    False Positives/Negatives: The sensor might produce false alarms or fail to detect gas leaks accurately if not properly calibrated, leading to potential safety concerns or unnecessary alerts.

2. Limited Detection Range

    Detection Range: The MQ-2 sensor has a limited detection range and may not be suitable for detecting very low or very high concentrations of gases, depending on the specific application and safety requirements.

    Specificity: The sensor is sensitive to a range of gases, which can make it challenging to distinguish between different types of gases or accurately measure specific concentrations without additional filtering or processing.

3. Environmental Interference

    Environmental Factors: Factors such as temperature, humidity, and air flow can impact the performance of the MQ-2 sensor. This can result in inconsistent readings and reduced reliability in varying environmental conditions.

4. Power Supply Dependence

    Power Requirements: Although the MQ-2 sensor and NodeMCU are relatively low power, the system still depends on a stable power supply. In cases of power failure or outages, the system may not function, potentially compromising safety.

CHAPTER 6

APPLICATIONS

1. Residential Safety

    Home Appliances: Monitors gas leaks from stoves, heaters, and other household appliances. The system alerts homeowners to potential gas leaks, reducing the risk of explosions or health hazards.

    Home Automation: Integrates with smart home systems to provide notifications through mobile apps or automated responses, such as shutting off gas valves or activating ventilation systems.

2. Industrial Facilities

    Manufacturing Plants: Detects gas leaks in industrial settings where flammable or toxic gases are used, preventing accidents and ensuring worker safety.

    Chemical Processing: Monitors environments where chemicals are stored or processed, providing early warnings to prevent leaks that could lead to dangerous chemical reactions or exposure.

3. Laboratories

    Research Facilities: Ensures safety in labs where gases are used in experiments or as part of the equipment. The system helps maintain a controlled and safe environment by detecting potential gas leaks promptly.

4. Automotive

    Vehicle Maintenance: Used in automotive workshops to detect fuel or exhaust gas leaks. It ensures that repairs are conducted safely and helps in maintaining vehicle safety standards.

5. Educational Projects

    Student Experiments: Serves as a practical project for students learning about electronics, sensors, and safety systems. It provides hands-on experience with real-world applications of technology.

6. Commercial Kitchens

    Restaurant Safety: Monitors gas lines and equipment in commercial kitchens, ensuring that any leaks are detected early to prevent potential fires or health hazards.

CHAPTER 7

CONCLUSION

In conclusion, the gas leak detection and alarm system utilizing an MQ-2 sensor and NodeMCU presents a robust and cost-effective solution for enhancing safety across diverse environments. By leveraging the MQ-2 sensor's ability to detect multiple gases and the NodeMCU's flexibility, the system provides reliable real-time monitoring and immediate alerts through visual and audible signals. The inclusion of **built-in Wi-Fi connectivity in the NodeMCU further enhances its functionality by enabling remote monitoring, data logging, and cloud-based notifications**, allowing users to receive alerts even when they are away from the site. This combination makes the system accessible for both hobbyists and professionals while offering broad potential for customization to meet specific needs.

Despite its advantages, the system does require careful calibration and may be affected by environmental factors, which can impact accuracy. Nonetheless, with proper implementation and maintenance, this setup effectively addresses gas leak detection challenges, contributing significantly to safety and risk management in residential, industrial, and educational settings.

CHAPTER 8

REFERENCES

1. **Espressif Systems**,
   ESP8266 NodeMCU Technical Reference Manual,
   Espressif Systems, 2018. https://www.espressif.com
2. **Blynk**,
   Blynk IoT Platform Documentation,
   Available: https://docs.blynk.io
   (Used for real-time gas monitoring through mobile application)
3. **Han, J., Kim, H., & Park, S.**,
   "IoT-Based Gas Leakage Detection and Alert System Using Wireless Sensor Networks,"
   International Journal of Engineering Research & Technology (IJERT), Vol. 8, No. 5, 2019.
4. **MQ-2 Gas Sensor Datasheet**,
   Hanwei Electronics,
   Used for detection of LPG, methane, smoke, and other combustible gases.

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